

Introduction to Aerial Robotics

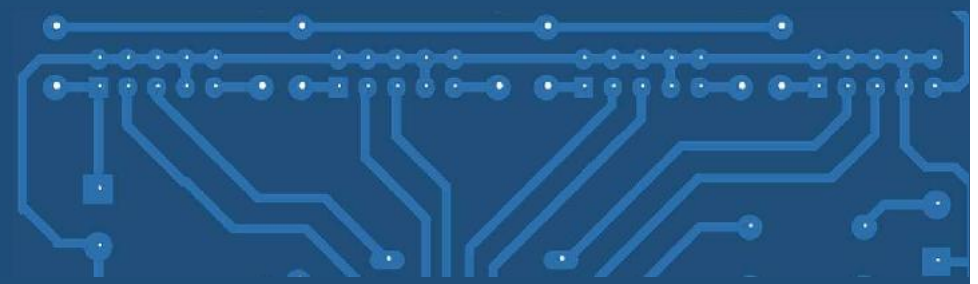
Part 2: Aircraft Parts, Forces and Flight Systems

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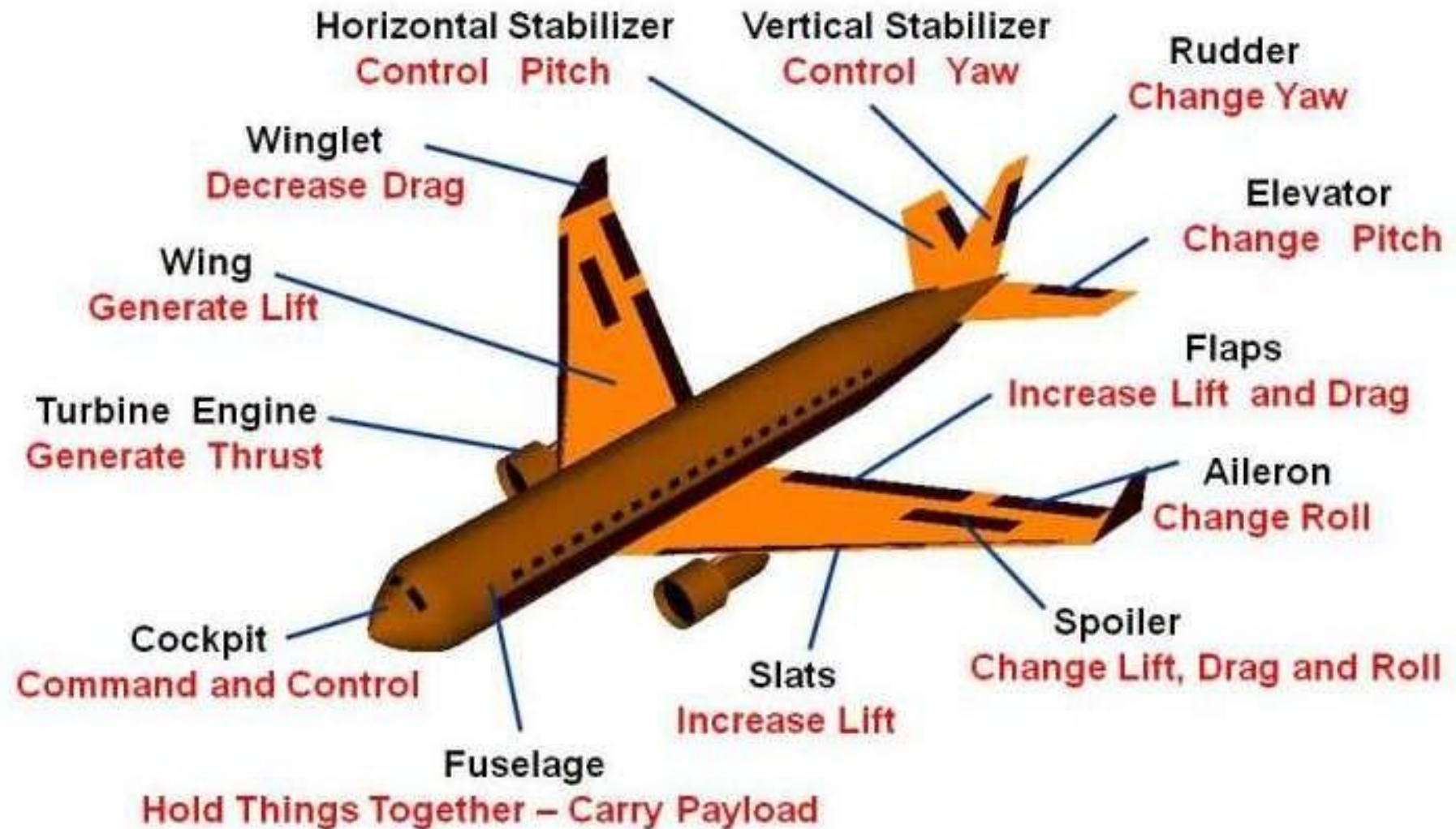
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Part 2 Learning Outcomes

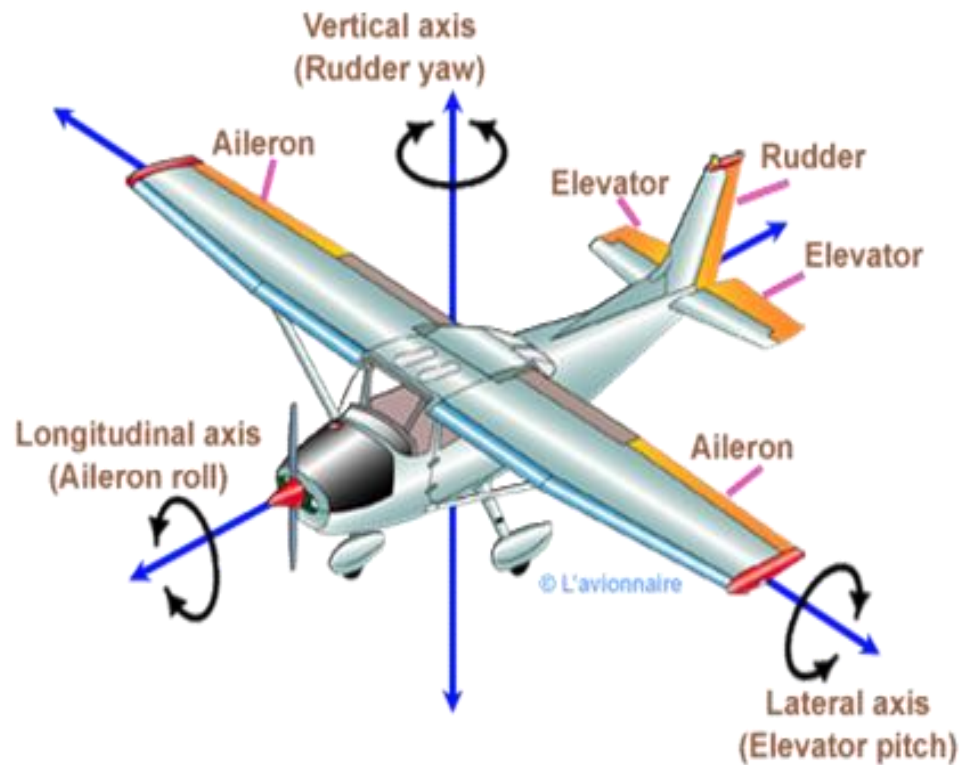
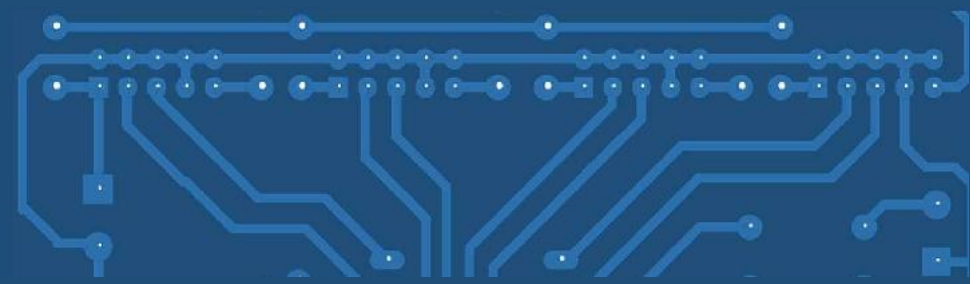


- Identify the main aircraft parts and explain their functions.
- Relate forces and moments to basic aircraft motion.
- Explain how fixed-wing aircraft, helicopters and quadrotors are controlled.
- Describe how power, sensing, communication and feedback support flight.

Fixed-Wing Aircraft Parts

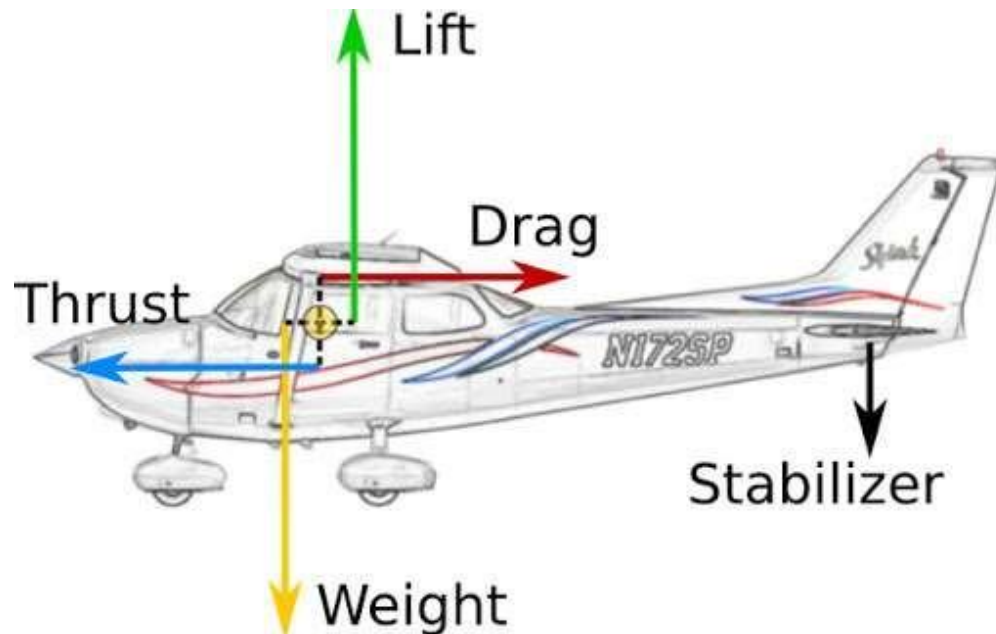


Fixed-Wing Aircraft Parts



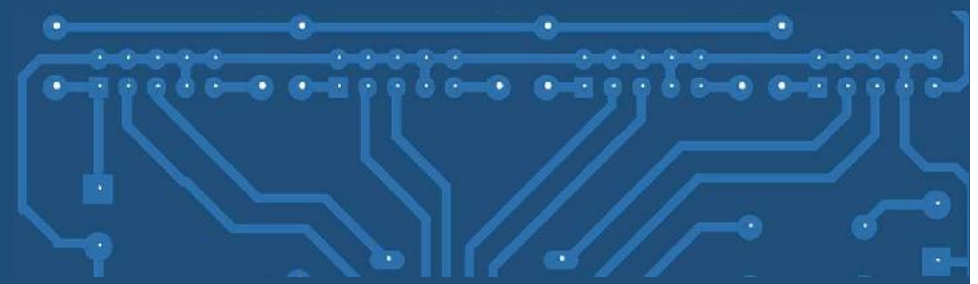
- Wings generate most of the lift.
- The fuselage carries the payload and onboard systems.
- Tail surfaces support stability and control.
- Ailerons, elevator and rudder change aerodynamic forces and moments.

Four Forces of Flight

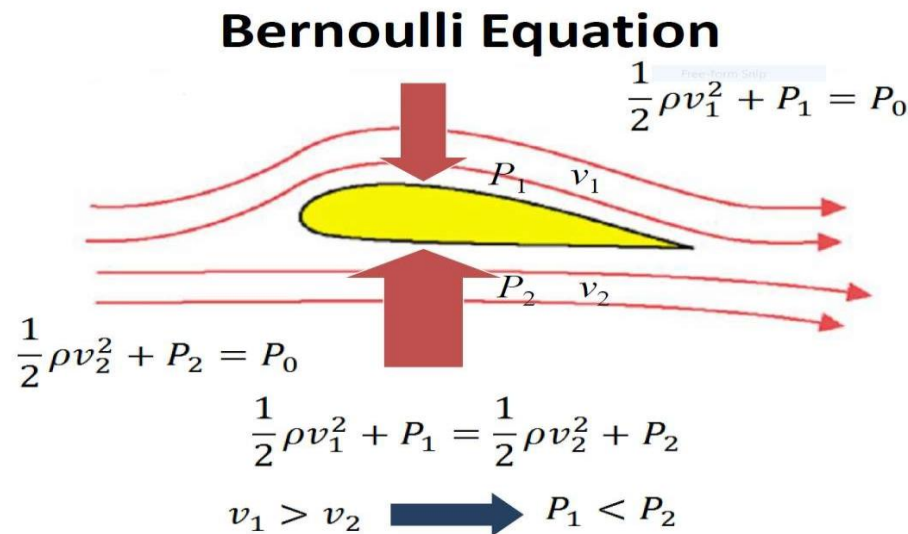


- Weight acts towards the Earth.
- Lift acts perpendicular to the relative airflow.
- Drag opposes motion through the air.
- Thrust comes from the propulsion system.

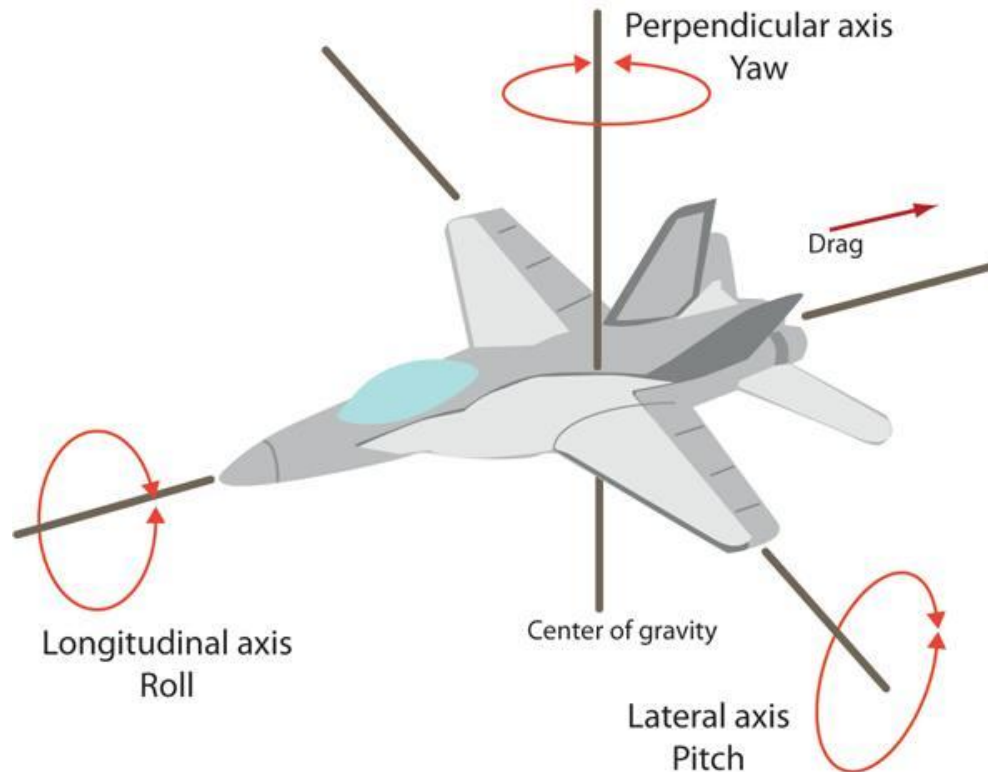
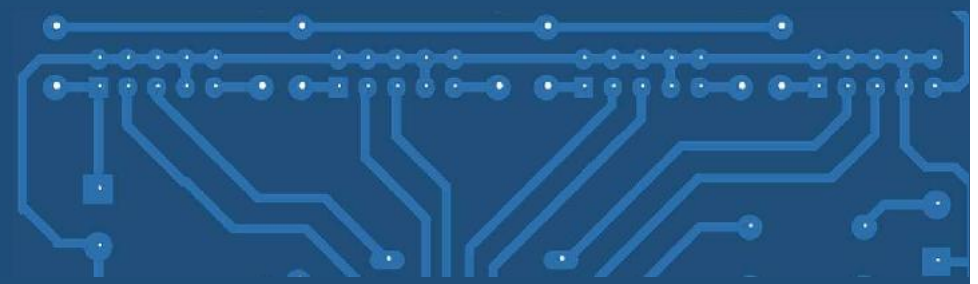
How a Wing Generates Lift



- A wing changes the surrounding airflow and deflects air.
- The pressure distribution over its surface produces an aerodynamic force.
- Lift depends on airspeed, air density, wing area and angle of attack.
- Exceeding the critical angle of attack can cause flow separation and stall.

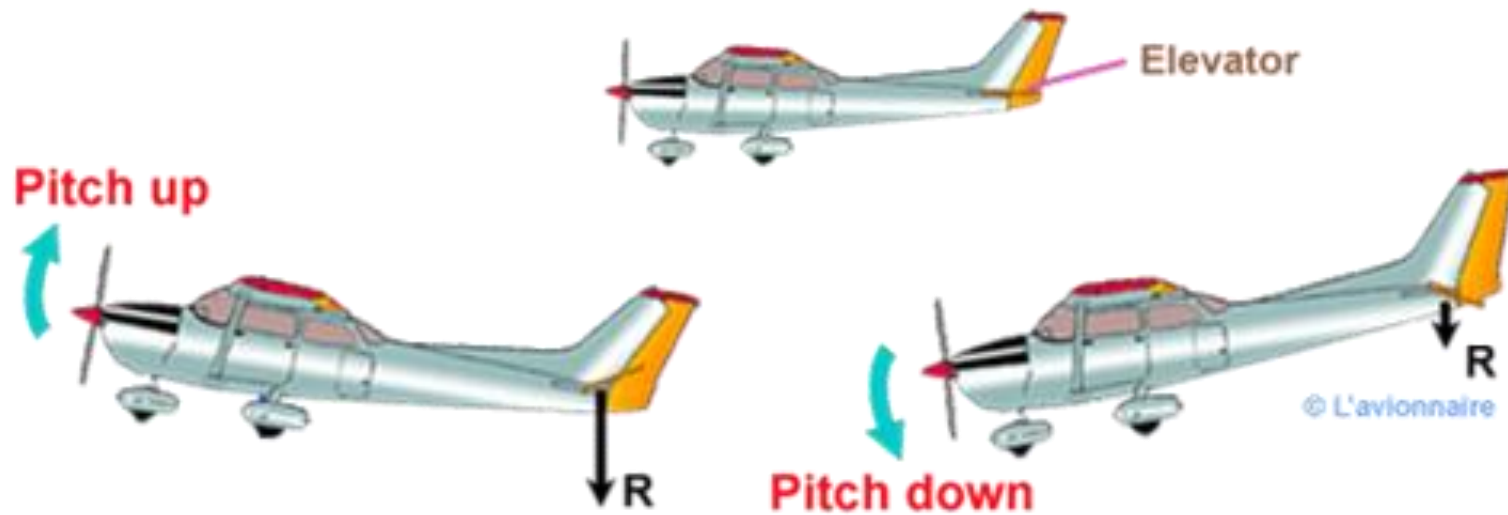


Aircraft Axes and Rotation



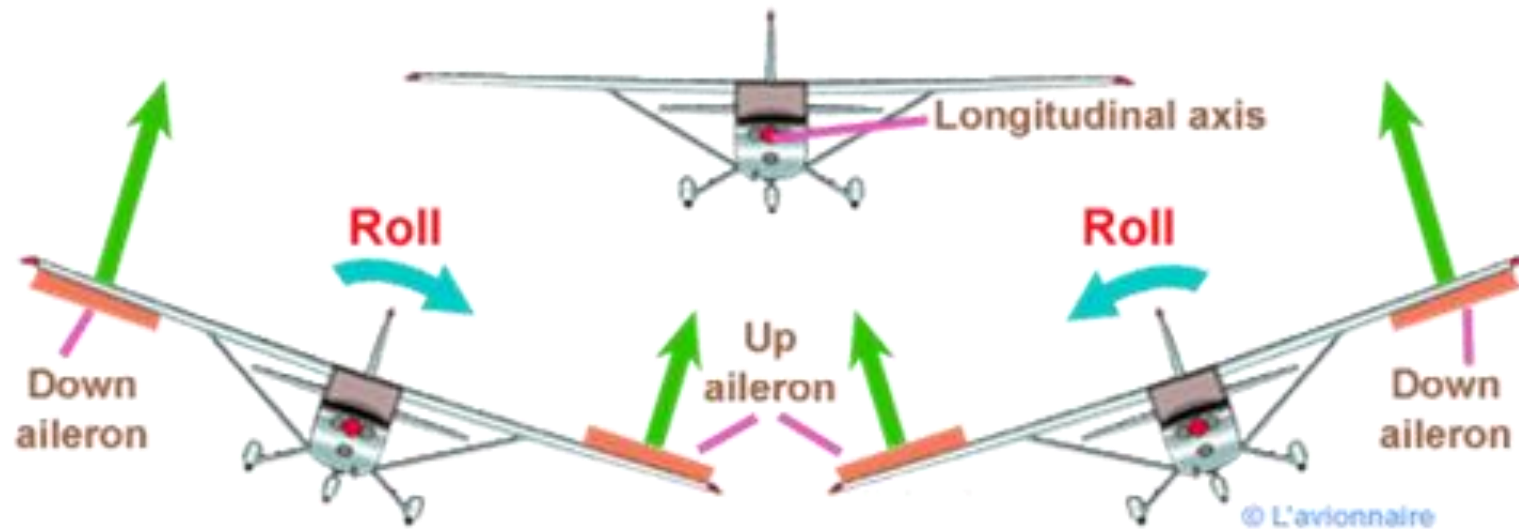
- Roll, ϕ : rotation about the longitudinal axis.
- Pitch, θ : rotation about the lateral axis.
- Yaw, ψ : rotation about the vertical body axis.
- Position describes location; attitude describes orientation.

Pitch Control



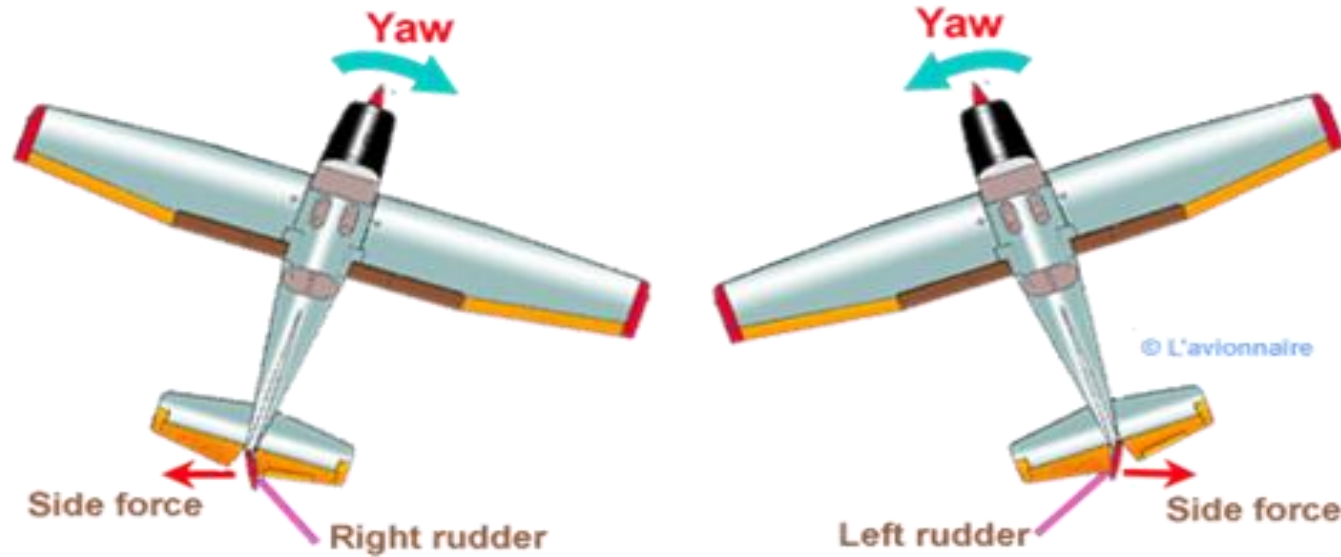
- The elevator changes the force on the horizontal tail.
- This creates a pitching moment about the centre of gravity.
- A nose-up or nose-down response changes the aircraft attitude.

Roll Control



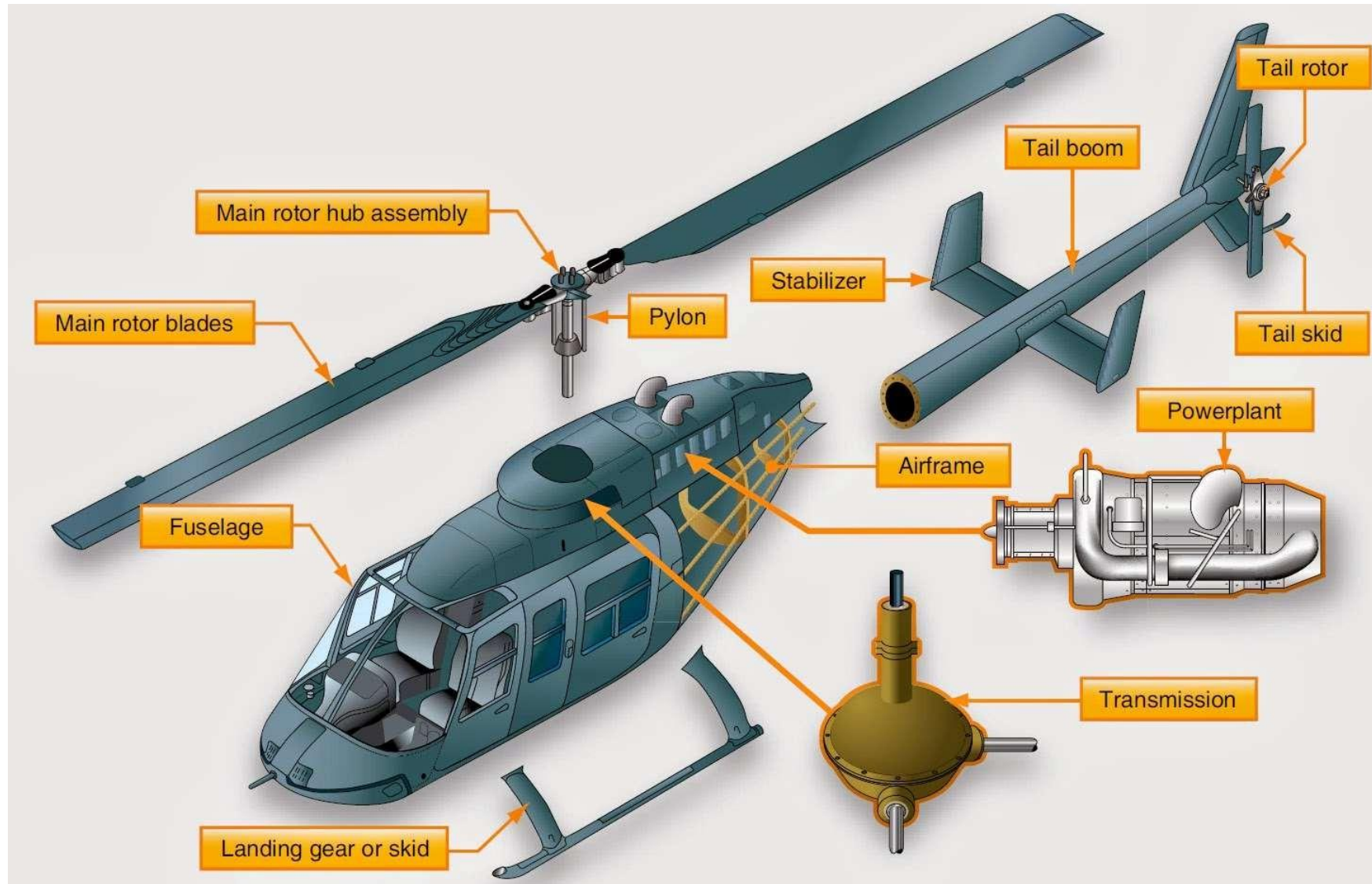
- Ailerons normally deflect in opposite directions.
- The resulting difference in wing lift creates a rolling moment.
- Banking tilts the lift force and supports a turn.

Yaw Control

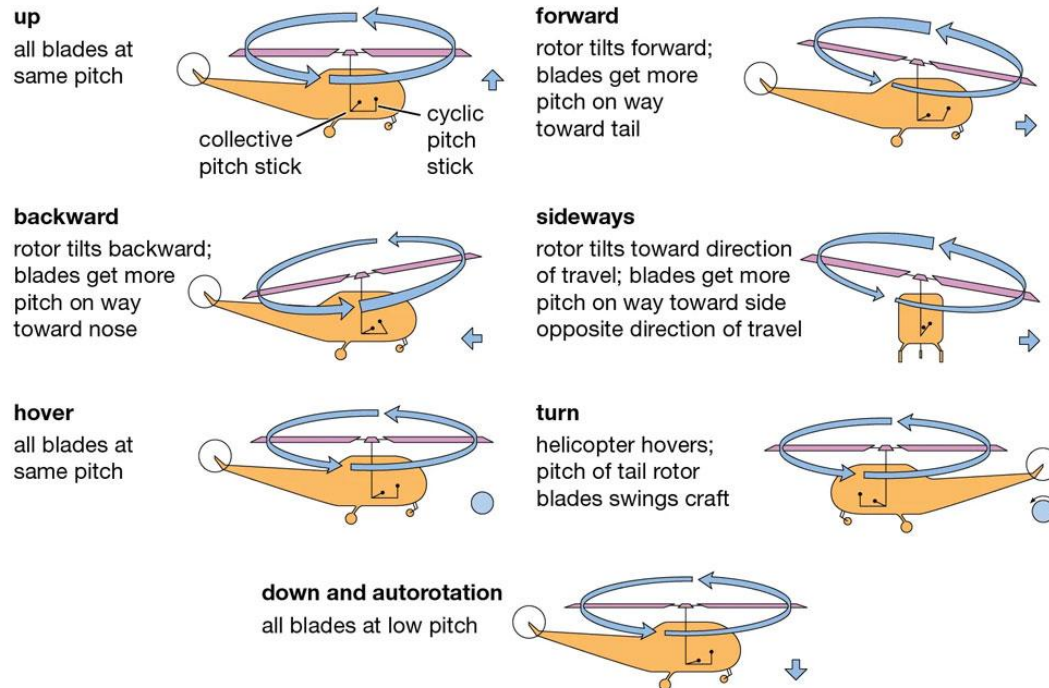


- The rudder changes the side force on the vertical tail.
- This creates a yawing moment about the centre of gravity.
- Rudder helps coordinate turns and manage sideslip.

Helicopter Parts



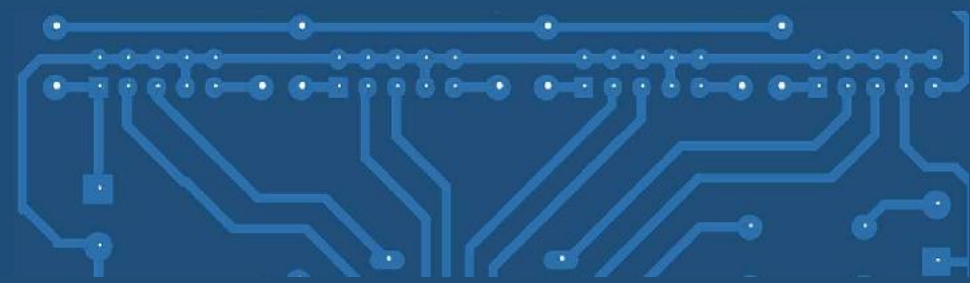
Helicopter Flight Control



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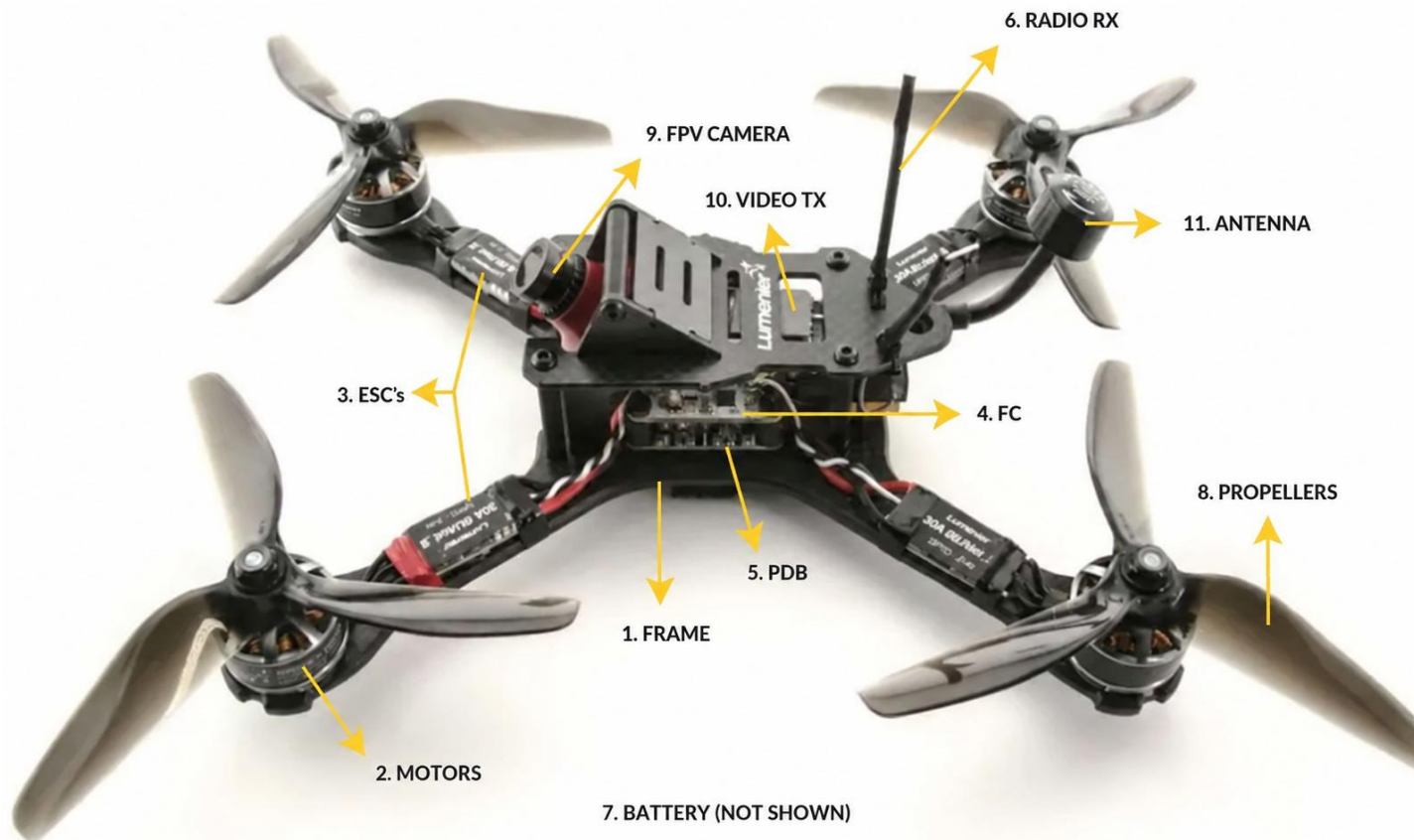
- Collective pitch changes all main-rotor blade pitches together.
- Cyclic pitch varies blade pitch around each rotation to tilt rotor thrust.
- The tail rotor counters main-rotor reaction torque and controls yaw.
- The swashplate transfers control inputs to rotating blades.

Helicopter Flight Control



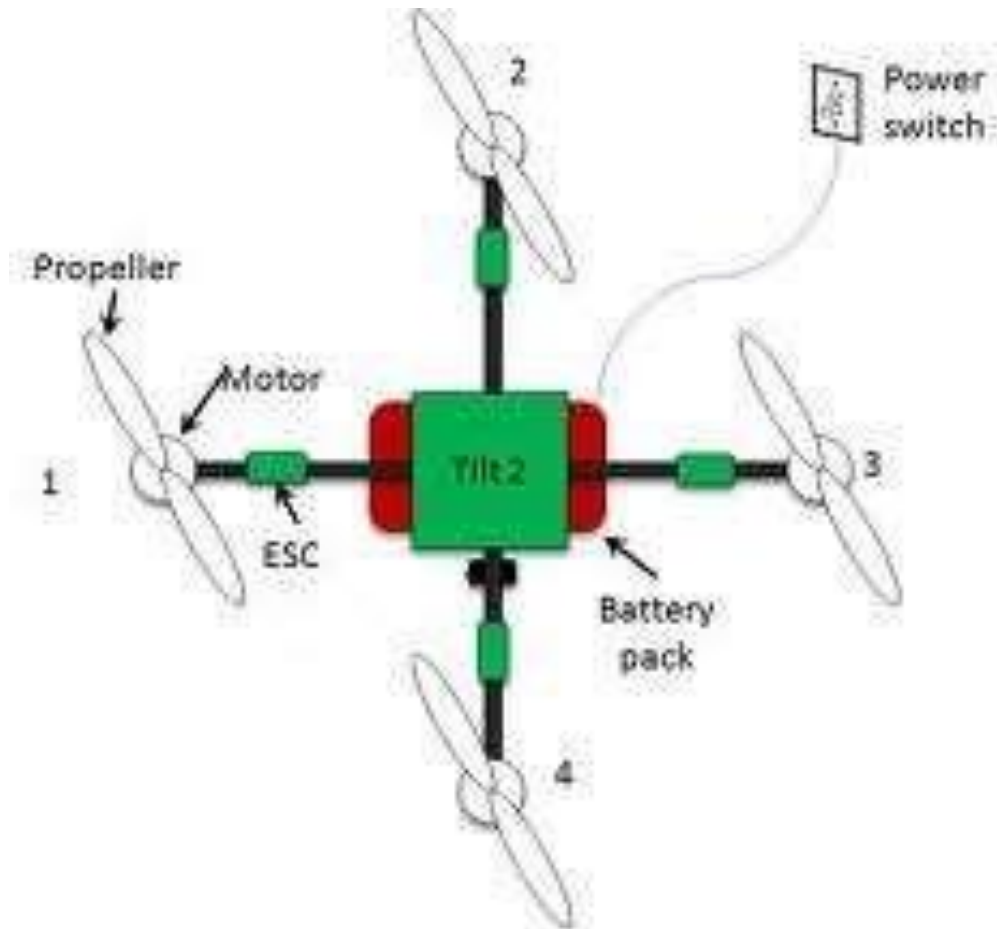
<https://www.youtube.com/watch?v=xairCL9m9ec>

Quadcopter Parts



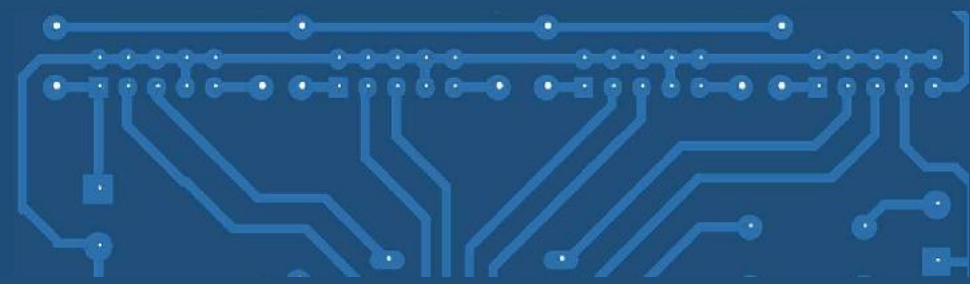
1. Frame
2. Brushless motors
3. Electronic speed controllers (ESCs)
4. Flight controller (FC)
5. Power distribution board (PDB)
6. Radio receiver (RX)
7. Battery (not shown)
8. Propellers
9. First-person-view (FPV) camera
10. Video transmitter (TX)
11. Video antenna

Quadrotor Parts and Actuation



- Four motors drive four propellers.
- The frame supports the battery, controller, sensors and payload.
- For a conventional fixed-pitch quadrotor, rotor speed changes thrust.
- Opposite rotation directions help balance reaction torques.

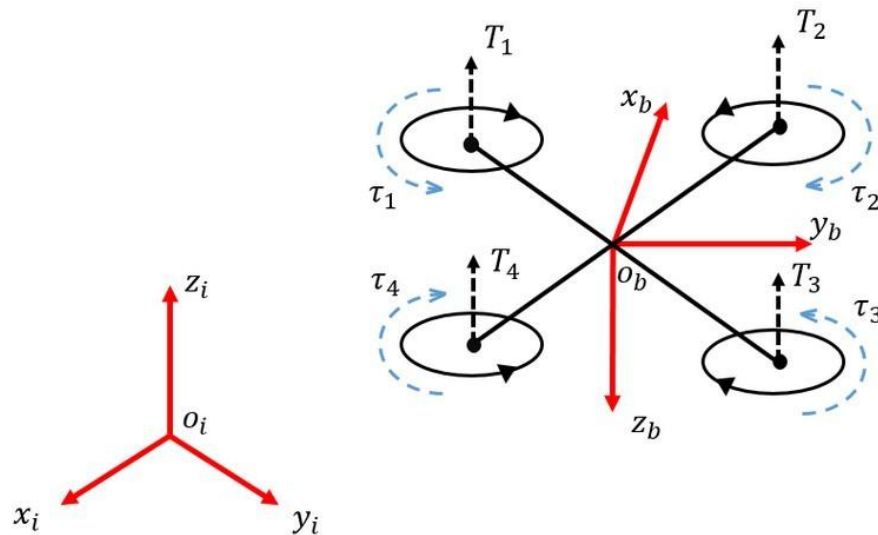
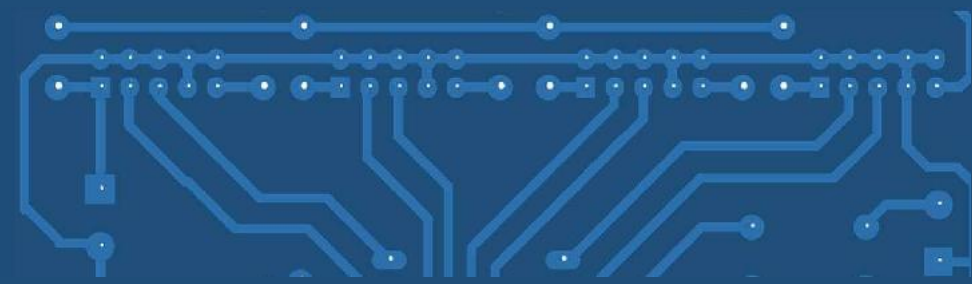
Hover and Vertical Motion



- In level hover, total rotor thrust balances weight.
- Increasing thrust above weight gives initial upward acceleration.
- Reducing thrust below weight gives initial downward acceleration.
- Constant vertical speed requires zero net vertical force.

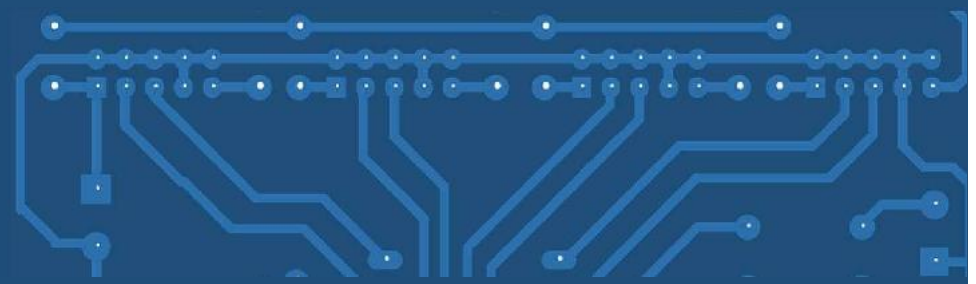
Level hover: $T_1 + T_2 + T_3 + T_4 = mg$

Quadrotor Roll and Pitch



- Unequal thrust across the aircraft creates a moment.
- A left - right thrust difference produces roll.
- A front–rear thrust difference produces pitch.
- The controller coordinates motor changes to manage total thrust.

Quadrotor Yaw



- Each rotor produces a reaction torque on the airframe.
- Balanced clockwise and anticlockwise rotor groups give little net yaw torque.
- Increasing one group and decreasing the other creates a yaw moment.
- Motor changes can approximately preserve total thrust.

Why a Quadrotor Tilts to Move

- Tilting directs part of the thrust horizontally.
- The horizontal component accelerates the aircraft.
- The vertical component must still balance weight to hold altitude.
- A tilted aircraft needs more total thrust for the same vertical support.

Constant altitude: $T \cos \alpha = mg$

Vertical Motion : Thrust and Weight

- Choose upward as positive for this one-dimensional model.
- Treat the aircraft as a point mass and neglect aerodynamic drag.
- Mass and applied thrust determine vertical acceleration.
- Full three-dimensional dynamics also include attitude, moments and inertia.

$$ma_z = T - mg$$

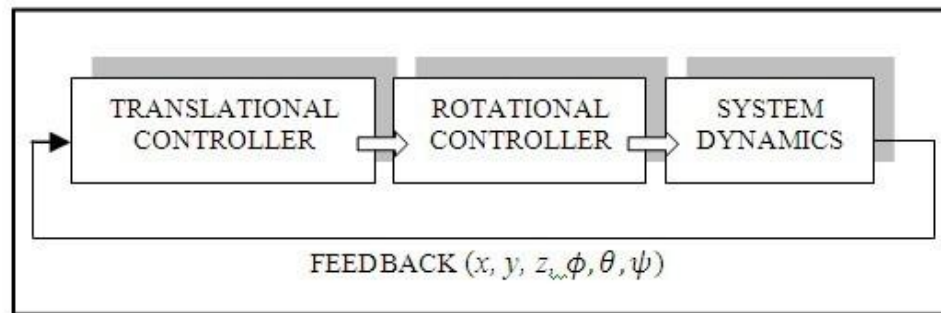
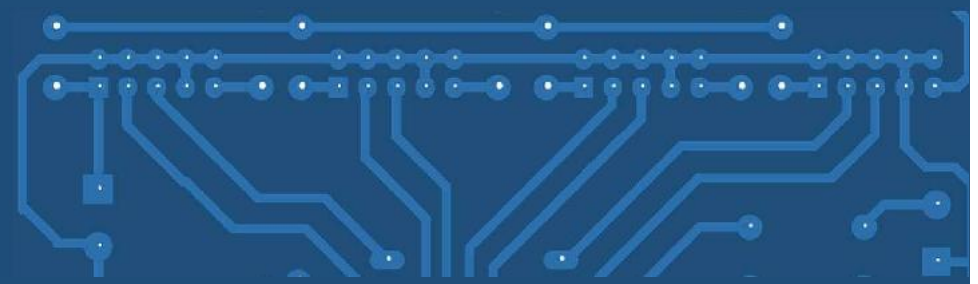
Worked Example: Hover Thrust

- A quadrotor has mass $m = 1.2 \text{ kg}$; take $g = 9.81 \text{ m/s}^2$.
- Required total thrust: $T = mg = 11.77 \text{ N}$.
- With equal rotor loading: thrust per rotor = 2.94 N .
- A 0.3 kg payload raises total hover thrust to 14.72 N , a 25% increase.

Quick Check : Vertical Acceleration

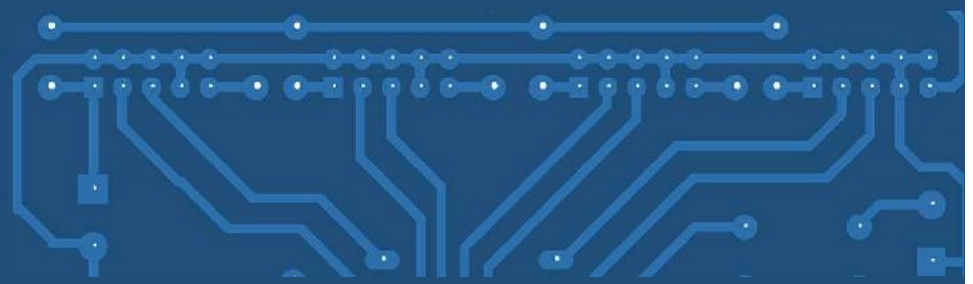
- A 1.5 kg quadrotor produces 18 N total thrust while level.
- Ignoring drag, calculate its initial vertical acceleration.
- Would keeping thrust at 18 N produce constant upward speed? Explain.

Flight Control and Feedback



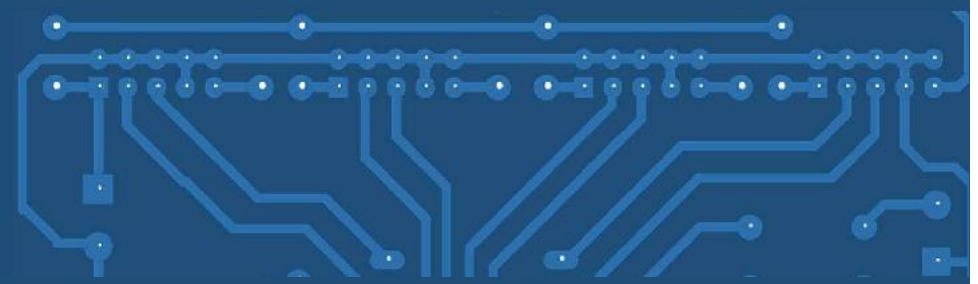
- A reference specifies the desired motion.
- Sensors and an estimator provide the current state estimate.
- The controller uses the error to command actuators.
- Feedback corrects deviations as the aircraft moves.

Disturbances and Limitations



- Wind and gusts change the forces on the aircraft.
- Payload changes affect mass, balance and required thrust.
- Sensor noise, bias and delay affect state estimates.
- Motor limits and battery voltage constrain control authority.

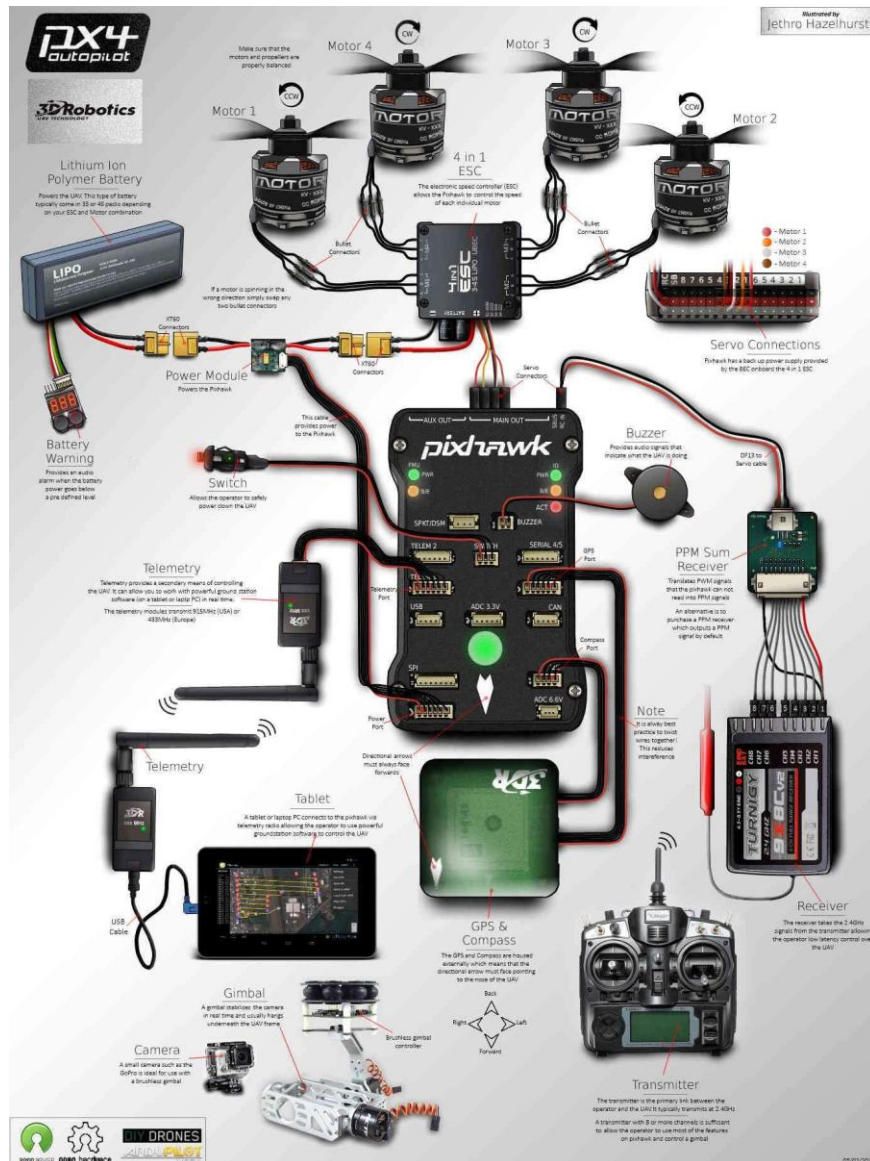
Frames and Configuration



- Frame geometry sets motor positions and moment arms.
- Payload placement affects the centre of gravity.
- Structural stiffness and vibration affect sensing and control.
- Extra rotors add actuators, mass and power demand; redundancy depends on design.

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Onboard Hardware



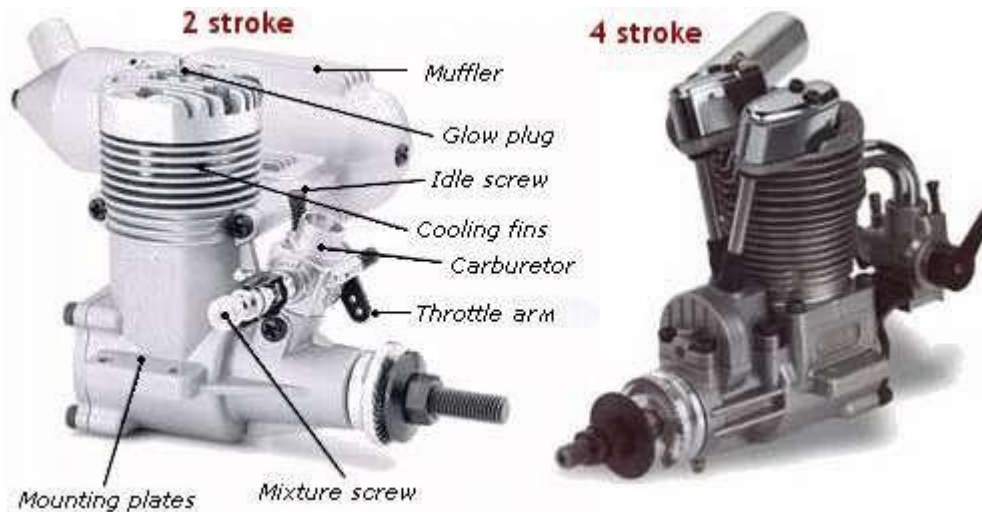
- Flight controller: runs estimation and flight-control tasks.
- Battery and power distribution: supply the electrical loads.
- ESCs and motors: convert commands and electrical energy into rotor motion.
- Sensors, radio links and payload complete the system.

Power Sources and Trade-offs

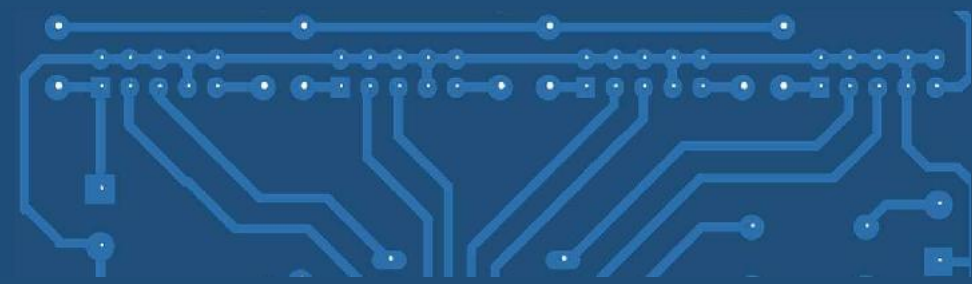
- Battery-electric systems offer straightforward motor actuation.

- Fuel engines need fuel storage, mechanical support and engine management.

- Mission duration, payload, noise and maintenance influence the choice.

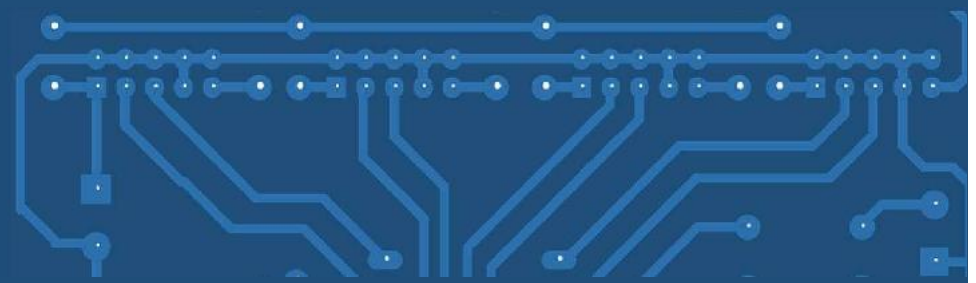


Reading a Battery Label



- Voltage indicates the electrical supply level.
- Capacity in ampere-hours describes stored charge.
- Approximate nominal energy is voltage multiplied by capacity.
- Discharge capability must match the current demanded by the propulsion system.

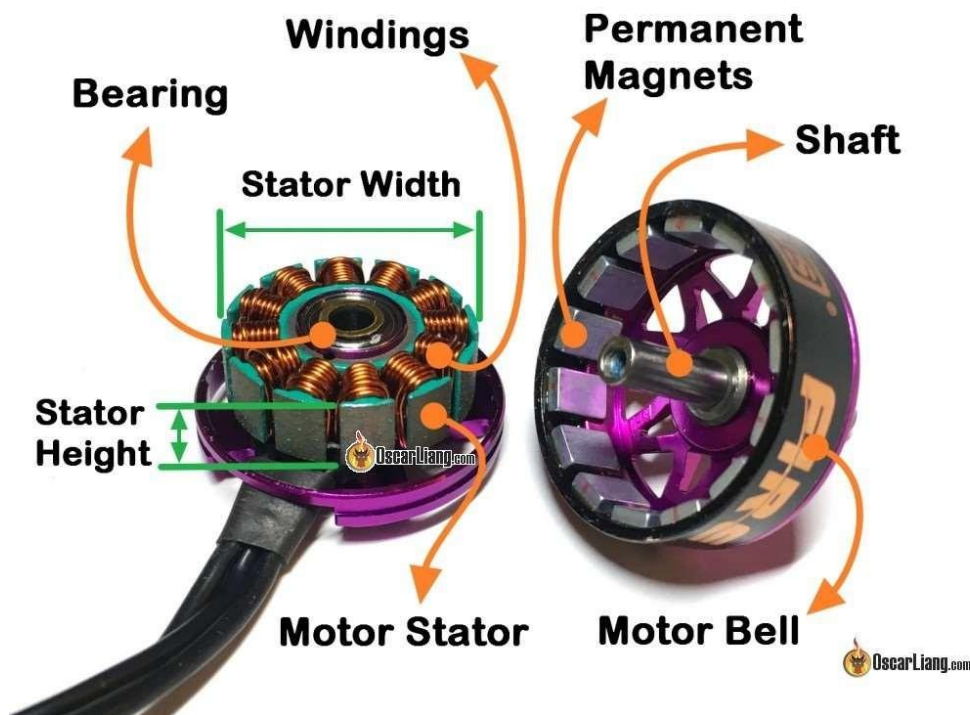
Estimating Endurance



- Example battery: 14.8 V and 5 Ah gives 74 Wh nominal energy.
- Assume 80% usable energy: 59.2 Wh.
- At an assumed average power of 250 W: $t \approx 0.237 \text{ h} \approx 14.2 \text{ min}$.
- Real endurance changes with flight condition, payload and battery condition.

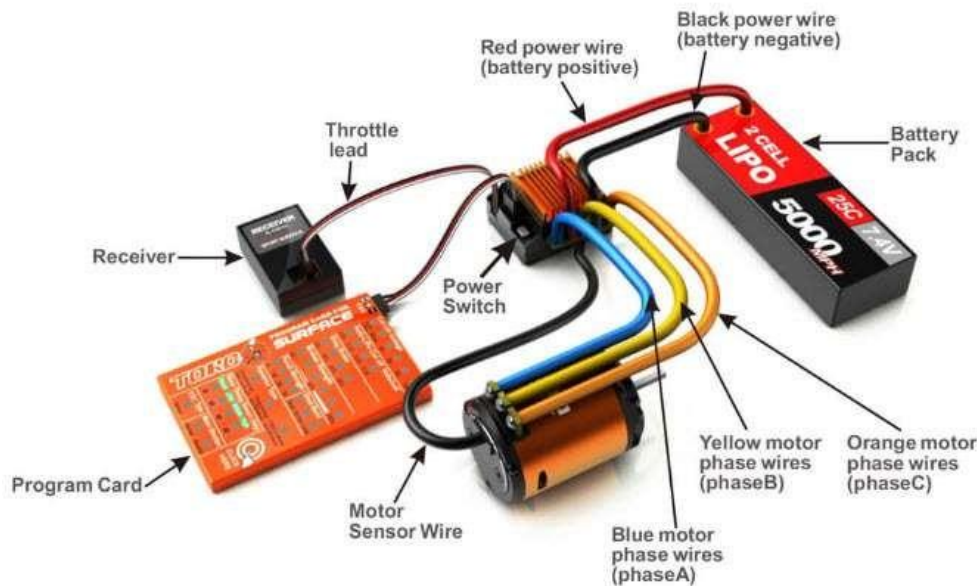
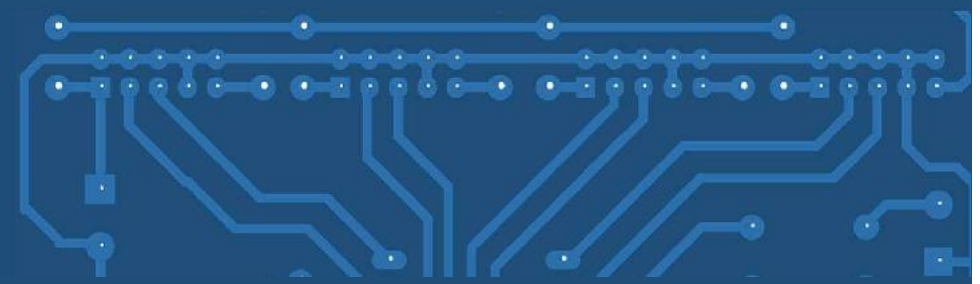
Estimated flight time $\approx$ usable energy / average power

Brushless Motors and Propellers



- Brushless motors use electronically controlled phase currents.
- The propeller converts shaft power into aerodynamic thrust.
- Motor, propeller and supply voltage must be selected together.
- Motor KV is approximately no-load speed per volt, not a power rating.

Electronic Speed Controllers



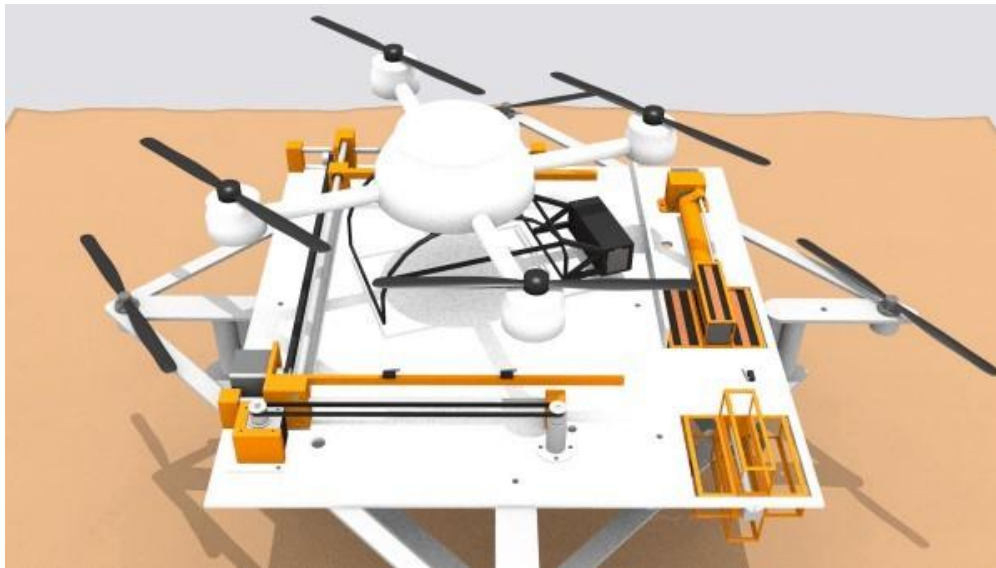
- An ESC switches electrical power to the motor phases.
- It receives commands from the flight controller.
- Its voltage and current ratings must suit the motor and battery.
- Rotor-speed response has a delay and affects control behaviour.

Matching Motor, Propeller and ESC



- Use thrust and current data for the actual motor - propeller combination.
- Check that the ESC and battery support the required voltage and current.
- Provide thrust margin above the hover requirement.
- Consider heating, vibration and operation across the battery voltage range.

Battery Swapping and Mission Time



<https://www.youtube.com/watch?v=geMQB2odVsA>

<https://www.youtube.com/watch?v=5efKmgCrL9k>

- Replacing a depleted battery can shorten turnaround time.
- A swap does not increase the endurance of one battery charge.
- Automated swapping adds docking, alignment and connection requirements.
- Assess the complete mission cycle, including landing and charging.

What Do Flight Sensors Measure?



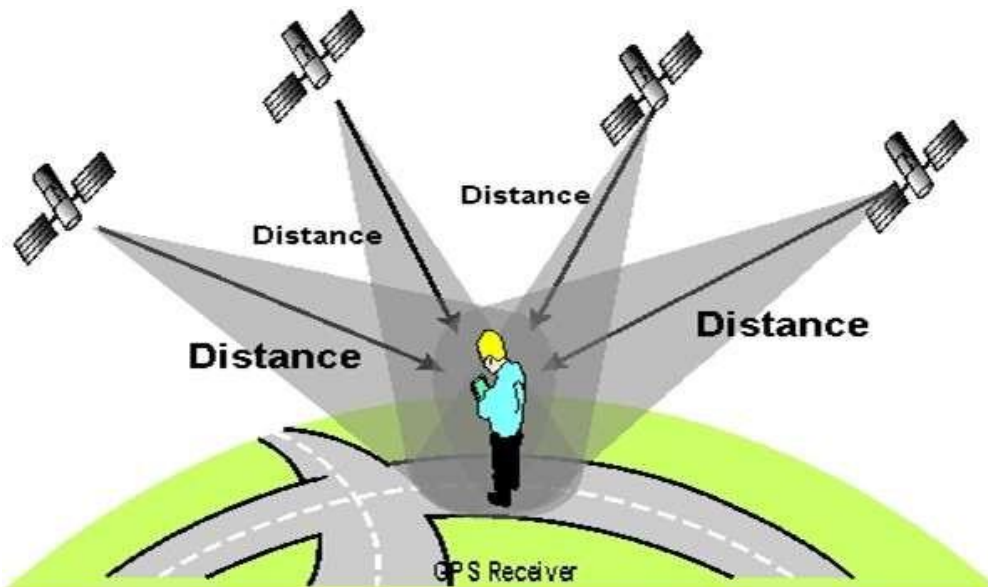
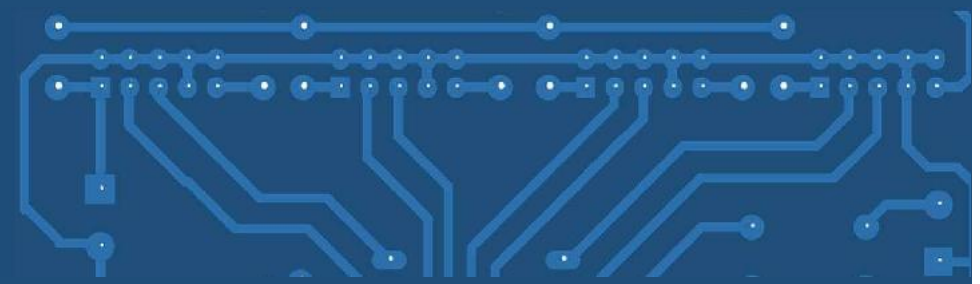
- Gyroscope: angular rate;
accelerometer: specific force.
- Magnetometer: magnetic field;
barometer: air pressure.
- GNSS receiver: position and velocity
estimates from satellite signals.
- Range sensors and cameras support
obstacle sensing and navigation.

The Inertial Measurement Unit



- An IMU typically combines three-axis accelerometers and gyroscopes.
- Accelerometers measure specific force, not position or velocity directly.
- Gyroscopes measure angular rate; integrating it estimates orientation change.
- Bias and noise cause drift, so other measurements are often combined.

GNSS Positioning



- GPS is one constellation within Global Navigation Satellite Systems.
- A receiver estimates position and often velocity using satellite signals.
- Accuracy depends on signal visibility, geometry, multipath and equipment.
- RTK and PPK use correction techniques; improved accuracy is conditional.

Why Combine Sensors?



- A gyro responds rapidly but accumulated bias causes orientation drift.
- Other measurements can constrain the drifting estimate.
- An estimator combines measurements with a motion model and uncertainty.
- The controller acts on the estimated state, not perfect knowledge of motion.

Telemetry and Ground Control



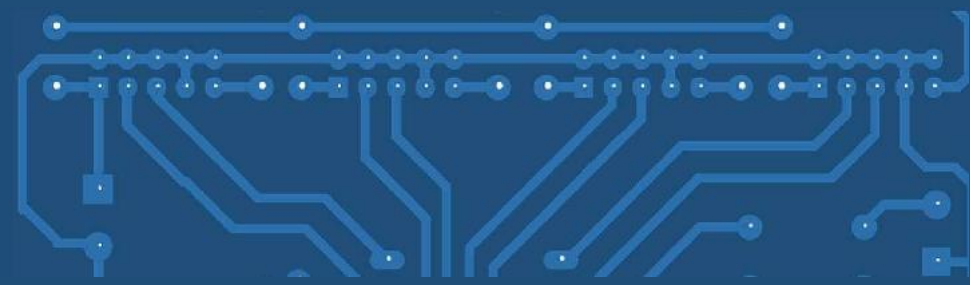
- Telemetry carries aircraft status, measurements and mission information.
- Command links send instructions from the ground system.
- Video links support observation or first-person-view operation.
- Range, latency, interference and link loss affect system design.

Onboard and Ground Processing

- Fast stabilisation loops run onboard to limit delay and link dependence.
- Onboard perception competes for electrical power, mass and cooling.
- Ground processing can support planning, supervision and data analysis.
- Choose processing location using latency, bandwidth and reliability needs.



Part 2 Summary



- Aircraft motion follows the balance of forces and moments.
- Control surfaces or rotor actuation generate the required moments.
- Power, structure and actuators constrain the achievable motion.
- Sensors, estimation and feedback support reliable flight.

Discussion: Explain the Flight System

- Why must a quadrotor increase thrust when it tilts while holding altitude?
- What changes when a payload is added?
- Why is a gyroscope alone insufficient for a long-term attitude estimate?
- Which computation must remain onboard if the radio link is delayed?